

Protein Screening Workflow & Scoring System

TE-Regulation Nuclear Protein Candidate Atlas

4,756
Input Genes

5,647
Reports Generated

4,128
Scored (Passed)

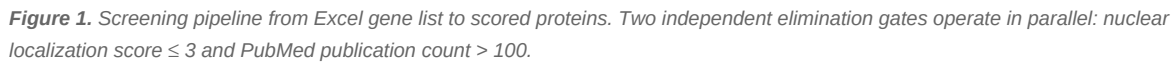
1,470
Eliminated

7
Categories

A systematic screening pipeline for identifying nuclear protein candidates relevant to transposable element (TE) regulation. Starting from 4,756 human genes, we applied a multi-dimensional scoring system across 7 subcellular localization categories to prioritize candidates for experimental validation.

Date: June 2026 · 4128 scored + 1470 eliminated

The screening process begins with all 4,756 genes from the TE-regulated protein finding Excel sheet. Each gene is evaluated across 7 dimensions and assigned to a subcellular localization category. Two elimination gates remove non-nuclear proteins and proteins with excessive literature coverage (PubMed > 100).



Gate	Criterion	Count	Description
Nuclear ≤ 3	■■■ score $\leq 3 / 10$	760	Mitochondrial, Golgi, ER, lysosomal, secreted, and other non-nuclear proteins. Nuclear score
PubMed > 100	PubMed publications > 100	710	Proteins with extensive existing literature are excluded due to low research novelty; the p
Scored	Passes both gates	4128	7 subcellular localization categories, ranked by composite score.

Table 1. Screening gates and their elimination criteria.

2. Multi-Dimensional Scoring System

Each candidate protein is scored on 6 core dimensions (0–10 scale) plus a cross-validation bonus. Dimensions are weighted to reflect their relative importance for TE regulation research. The final normalized score (0–100) enables direct comparison across all categories.

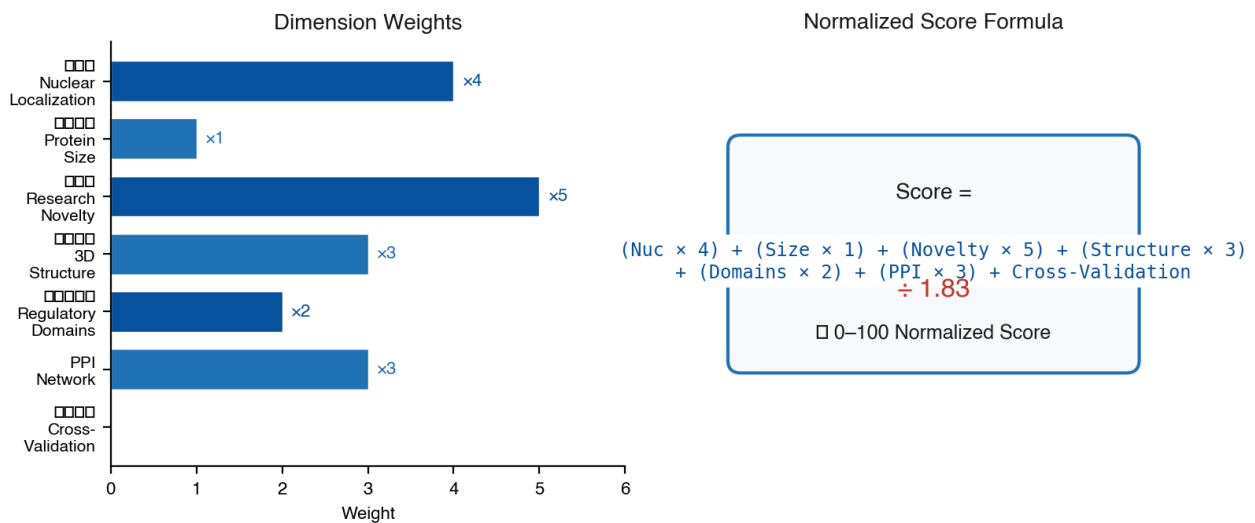


Figure 2. (Left) Dimension weight distribution. Nuclear localization (×4) and research novelty (×5) carry the highest weights, reflecting their primacy in candidate prioritization. (Right) Normalization formula. The divisor 1.83 maps the theoretical maximum of 183 to a 0–100 scale.

Dimension Definitions

Dimension	CN	Weight	Data Sources
Nuclear Localization	■■■■■■■	×4	UniProt subcellular location, HPA IF images, GO cellular component (IDA/IMP)
Protein Size	■■■■■	×1	Amino acid length, molecular weight (kDa). Optimal range: 300–800 aa
Research Novelty	■■■■■■■	×5	PubMed publication count (favoring < 50), targeted literature search relevance
3D Structure	■■■■■	×3	AlphaFold pLDDT, PDB entries, domain-level structural confidence
Regulatory Domains	■■■■■■■	×2	InterPro/SMART domain annotation, chromatin/DNA-binding domains, enzymatic domains
PPI Network	PPI ■■■	×3	IntAct, STRING, BioGRID interaction data. Weighted by regulatory partner enrichment
Cross-Validation	■■■■■	bonus	Concordance across databases (0 to +3). Rewards multi-source agreement

Table 2. Scoring dimensions with data sources and weight justification.

3. Score Distribution Analysis

The overall score distribution and per-category breakdown reveal the statistical properties of the scoring system. The boxplots enable direct comparison of score medians, spreads, and outliers across subcellular localization categories.

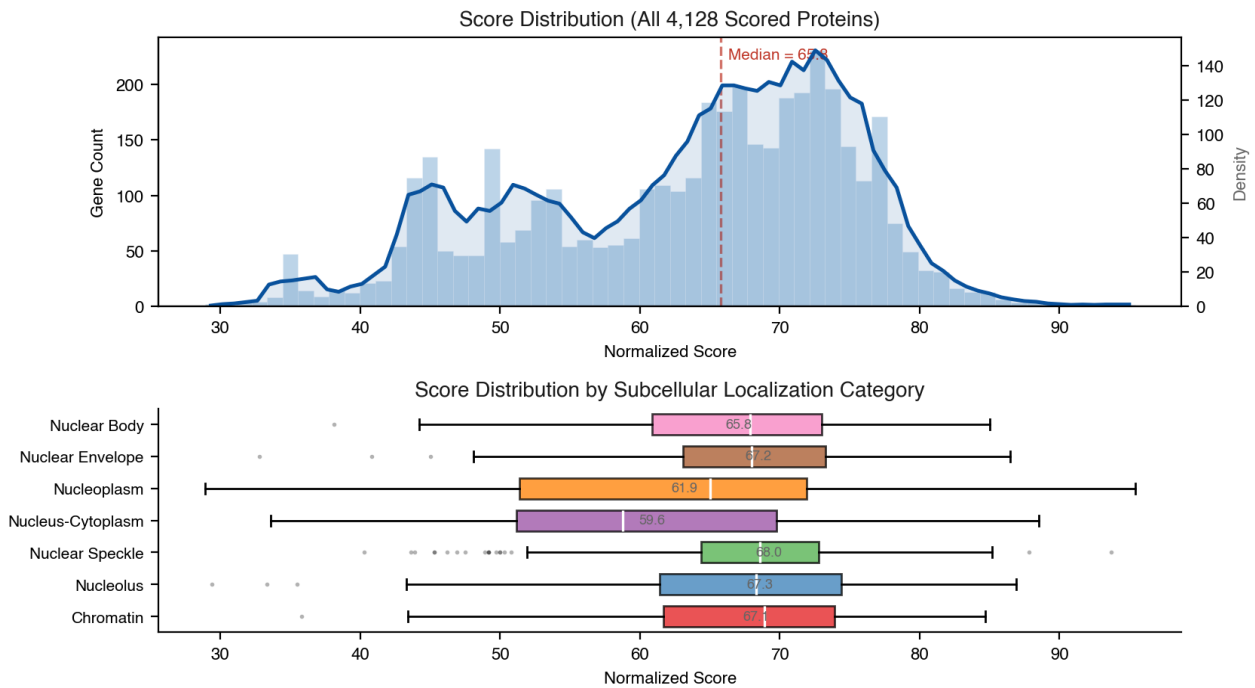


Figure 3. (Top) Overall normalized score histogram with kernel density estimate overlay. Dashed red line indicates the median. (Bottom) Boxplot comparison across 7 subcellular localization categories, showing distribution spread, quartiles, and outlier patterns.

4. Category Composition

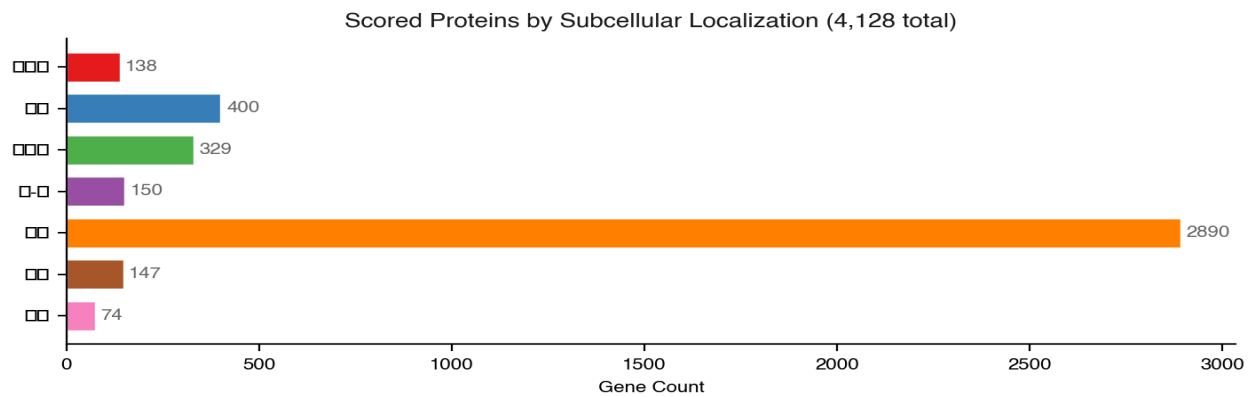


Figure 4. Breakdown of 4,128 scored proteins across 7 subcellular localization categories. Nucleoplasm is the largest category (2,890 proteins), followed by Nucleolus (400), Nuclear Speckle (329), and Chromatin (138).

5. Category Phenotypic Fingerprints

Median dimension scores reveal distinct phenotypic profiles for each subcellular localization category. These 'fingerprints' help identify category-level biases — for instance, Chromatin proteins tend to score higher on regulatory domains, while Nucleolus proteins show stronger PPI networks.

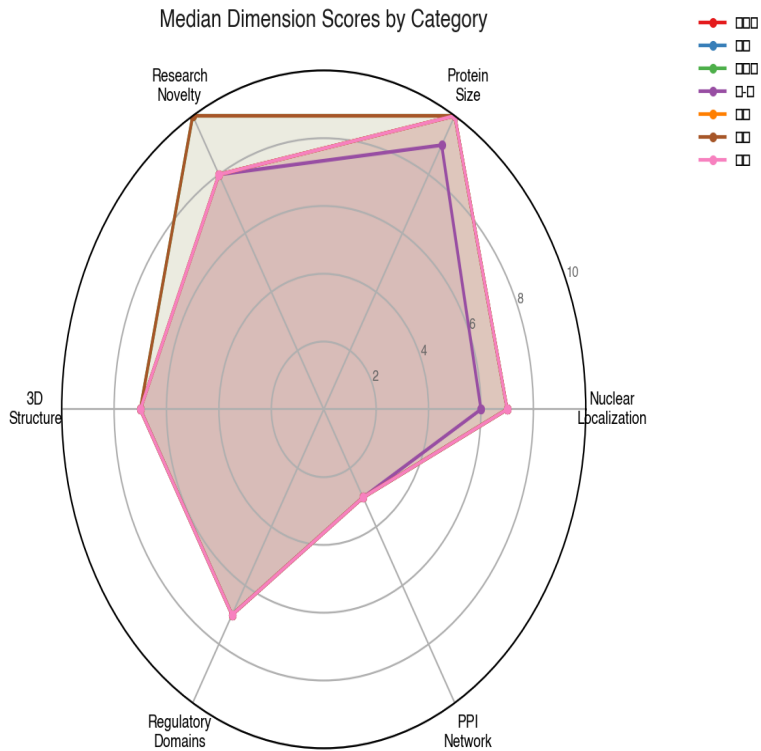


Figure 5. Radar chart of median dimension scores (0–10) per category. Each axis represents one scoring dimension. Chromatin proteins (red) show elevated nuclear localization and regulatory domain scores. Nuclear Speckle proteins (green) show higher PPI network richness.

Star Rating Key

Score Range	Stars	Interpretation
85–100	★★★★★	Exceptional candidate — top priority for experimental validation
75–84	★★★★	Strong candidate — well-supported across most dimensions
65–74	★★★	Good candidate — solid scores with some dimension gaps
55–64	★★	Moderate candidate — merits further investigation
< 55	★	Baseline candidate — review before prioritization

Table 3. Star rating system for candidate prioritization.